Background Knowledge in Random Forest Reconstruction: Evidence, Baselines and Evaluation Gaps

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Abstract—Releasing a trained random forest can reveal training data, but reconstruction risk depends on the attacker's background knowledge. This review compares eight studies on reconstruction, attribute inference and evaluation. The evidence highlights the distinction between compatible tables and original records, and the importance of informed baselines and row matching. The reviewed joint forest studies leave open how the identity of a known feature affects recovery of a fixed target. We propose comparing two known features while keeping the bagged, non-differentially private forest, target and solver budget fixed, with a model-free control and matching sensitivity check. The review synthesizes reported evidence; no new experiments were run.

1 Introduction

Releasing a trained random forest (RF) model can be dangerous, because information about the training data can be extracted from the tree splits and per-class leaf sizes of the records in overlapping groups of features. An attacker can even reconstruct the entire training table in full detail that was not released to the model author, in critical domains such as income, credit, and criminal justice. How much does the model contribute to what the attacker already knows?

Feature domains (and their corresponding values for selected training features), background knowledge in the form of auxiliary population data, and background knowledge in the form of aggregate relationships (e.g. fairness) all provide different information to the attacker. That is, attribute inference on a known individual, finding a compatible data table, and joint reconstruction of original training records all are different tasks [1], [2], [3].

This review therefore focusses on the impact of knowledge about the attack on the interpretation of the reconstruction error for the joint reconstruction of tabular data from RFs. We will review seven studies plus one older study on tree inversion in detail. The main topics to be discussed are 1) how model information is used by the attacks, 2) what background knowledge adds and 3) how success of reconstruction is measured. For attribute inference as well as for images suitable controls and a measurement method will be picked from the relevant studies and applied to the model settings that are studied in these papers. The niche between the two studies on the proposed known feature(s) vs. the target feature in a fixed bagged non-DP forest will be investigated in detail.

2 Method

The review covers eight scientific papers on background knowledge in reconstruction attacks. It began with two papers already available: *Trained Random Forests Completely Reveal your Dataset* [4] and *Probabilistic Dataset Reconstruction from Interpretable Models* [2]. Keyword searches and reference lists provided further candidates. Since the topic is still emerging, the search gave priority to recent studies. Screening these candidates helped narrow the research question and the review's scope.

2.1 Search sources and keywords

The search used Google Scholar and DBLP, combining keyword queries with reference lists. Queries covered tree reconstruction, attacker knowledge, model-free baselines and evaluation. For tree reconstruction, they included *random forest*, *dataset reconstruction*, *differential privacy* and *training set reconstruction*. For attribute inference, they included *inference*, *attribute inference*, *feature inference*, *attribute inference attacks*, *sensitive attributes* and *private sensitive attributes*. Evaluation queries included *reconstruction attacks*

and *reconstruction attacks metrics*. These searches identified the studies by Gorgé et al. [5], Mehnaz et al. [6], Jayaraman and Evans [7] and Wen et al. [3].

The reference list of the probabilistic reconstruction paper [2] led to the fairness study [8]. References in Mehnaz, the fairness study and the SoK led to Fredrikson et al. [1]. Mehnaz's study was first found through search; its citation in Jayaraman and Evans further supported its relevance.

Publication details were checked against DBLP records and their BibTeX entries, ACM, IEEE, PMLR and USENIX records, and the available manuscripts. The selected papers come from ICML, ACM CCS, IEEE SaTML and USENIX Security. This focused search does not provide exhaustive coverage.

2.2 Eligibility, screening and extraction

To be included, papers needed an available full text and verified publication or acceptance at a well-established, peer-reviewed conference or journal. They also had to cover at least one of three areas: reconstruction from tabular or interpretable models; attacker knowledge and model-free comparisons; or definitions and evaluation of reconstruction success. Fredrikson et al. [1] provide tree-inversion experiments, while Wen et al. [3] contribute an original benchmark and metric analysis. Studies of other model families were included when they addressed these methodological questions. The review excludes membership-only attacks, model extraction, synthetic-data utility without recovery of original records, and image inversion without a relevant evaluation contribution. Code availability was useful, but not required.

Screening titles and abstracts for relevance to reconstruction attacks and background knowledge helped refine the question. Candidates were then checked against the inclusion and exclusion criteria. The final selection brings together studies of random forest reconstruction and methodological work on attacker knowledge, model-free baselines and evaluation. These contributions form the review's three themes.

For each paper, the information extracted covered the task, model access, prior information, datasets, method, metrics, baselines, computational budget, results and limitations. Table 1 and the synthesis compare these elements, referring to source sections, tables and appendices where needed. Results and specific passages are interpreted with attention to whether the source is a published PDF or an author manuscript. The review uses reported evidence; no code was run and no experiments were repeated. Source-version differences and limited search coverage constrain its conclusions and any claims of novelty.

3 Synthesis of findings

Table 1 compares model information, prior knowledge and targets. The studies span individual inference, joint reconstruction and evaluation. Differences in their tasks and assumptions prevent a direct ranking of results.

3.1 From individual inversion to joint constraints

Early tree inversion estimates a missing attribute for a known individual. Fredrikson et al. [1] combine prediction confidence, a confusion matrix and marginal statistics in a maximum-a-posteriori inverter. With white-box access, the inverter also uses paths and counts; additional unknown attributes can be marginalized (Section 4). Mehnaz et al. [6] later show that labels alone can support attribute inference. They test possible sensitive values against the known true label and use selected attack records to train a predictor. Their Adult decision-tree experiments report roughly 70% accuracy, compared with roughly 61% for the earlier method (Appendix A.2, Table 13). Although the attack uses less output information, it still requires substantial knowledge of each record and infers attributes of known individuals. Measuring the model's added value therefore requires an informed no-model predictor.

Joint reconstruction uses tree information to describe or recover a whole table. Ferry et al. [2] describe distributions over compatible tables, including dependencies between cells. For example, rejecting a conjunction in a rule list can rule out combinations without ruling out their individual values. Joint uncertainty captures this information, which cellwise models miss. Under uniform-world and row-independence assumptions, tree entropy can be calculated using leaf supports. For rule lists, antecedents can overlap, and the direct entropy calculation accounting for that overlap has factorial cost in the number of rules (Section IV-C, Equation 16).

DRAFT [4] finds a table that satisfies constraints across several trees. Reconstructed features and labels must follow the observed paths and reproduce class counts. Bagging makes sampling multiplicities unknown, and DRAFT handles them with a likelihood objective. The main experiments use binarized

Table 1
The selected studies in selection order. Each row describes the relevant setting. Information and computational budgets differ across studies.

Study	Model observation	Prior information	Target / measure
DRAFT, 2024 [4]	RF structure and class counts	Size, domains, encoding; exact columns in ablations	Joint table; matched bit error
Probabilistic, 2024 [2]	Trees/rule lists and supports	Size and domains in evaluated applications	Possible tables; entropy ratio
DRAFT-DP, 2026 [5]	Random-split RF, noisy counts	Mechanism/domains; columns or records varied	Table; error and exact rows
Mehnaz, 2022 [6]	Classifier scores or labels	Identified records, other attributes, true label	Sensitive attribute; accuracy/MCC
Jayaraman and Evans, 2022 [7]	Neural outputs or internal signals	Other candidate attributes; auxiliary data	Attribute inference; top-100 PPV
Fairness, 2023 [8]	Predictions, fairness relationship	Known non-sensitive rows, labels, auxiliary data	Sensitive column; accuracy
SoK, 2025 [3]	Access varies across benchmark attacks	Auxiliary data vary across attacks	Mainly images; distance/coverage
Fredrikson, 2015 [1]	Tree confidence/structure/counts	Known attributes, true label, marginal statistics	Individual attribute; precision/recall

Adult, COMPAS and Default Credit samples of 100 records, five seeds and a five-hour solver limit. Deep forests without bagging can give zero matched error in successful runs. With bagging, reconstruction becomes more ambiguous and computationally difficult; some unsuccessful solves are excluded from the reported averages. DRAFT supports practical joint reconstruction from a released RF, although a forest can be consistent with more than one training table.

Noisy counts can violate DRAFT's exact-count constraints. DRAFT-DP [5] handles this problem by jointly estimating records, underlying counts and likely noise. Its forest uses data-independent random splits, no bootstrap and leaf counts with Laplace noise. With ten trees of depth five and $\varepsilon = 5$, reported matched bit errors are about 0.12, 0.06 and 0.16 on Adult, COMPAS and Default Credit, compared with 0.25, 0.21 and 0.26 for the random baseline (Table I). These results are compatible with DP's guarantee about participation. Figure 2 compares DRAFT on the same forests with unperturbed counts, isolating count perturbation within that design. Across the two papers, however, the known-column experiments use different forest configurations.

DRAFT-DP extends the constraint formulation to a noisy release, but the approaches above address different targets and assumptions. To compare the contribution of background knowledge, model access and observations should remain fixed.

3.2 What knowledge contributes beyond the release

Exact columns and auxiliary population data provide different information. In DRAFT's Appendix G [4], the attacker knows selected original features for every training row, including all their encoding bits. More known columns generally reduce error on the remaining attributes in the tested bagged forests with 100 trees and unrestricted depth. As columns become known, the remaining attributes change too. The experiment thus varies both attacker knowledge and the target set used for scoring.

DRAFT-DP's Appendix F [5] reports no improvement from additional known columns with ten trees, depth five and $\varepsilon = 5$. The authors suggest row alignment as an explanation for a slight error increase. The result differs from DRAFT's, yet the settings also differ in splitting rules, bootstrap, depth, ensemble size, noise and budgets. Comparing them cannot isolate how DP affects the value of known columns. Neither experiment compares two specific known features against one fixed target.

The identity of known attributes is already part of the study by Mehnaz et al. [6]. They vary the missing non-sensitive attributes while keeping Adult marital status and the decision tree fixed (Appendix A.3, Figure 13). Precision, recall, G-mean and MCC stay broadly stable in the tested cases. Query costs can increase sharply with the domains of the missing

features. Our proposed RF comparison extends this question to one known column and anonymous joint reconstruction.

Evaluating inference with restricted model outputs also requires an informed baseline. Jayaraman and Evans [7] compare neural attribute-inference attacks with imputation from auxiliary data without the target model. On Texas-100X with 500 same-distribution auxiliary records, Table 4 reports top-100 positive predictive value of 0.39 ± 0.03 for imputation and 0.52 ± 0.03 for the white-box attack. With more auxiliary data, imputation can become stronger, so relative performance depends on the setting. For RF reconstruction, this motivates testing what the model adds to existing knowledge. The reported numbers themselves concern neural models, known candidates and selected-candidate PPV.

Fairness correction [8] shows how an aggregate relationship can add information. An auxiliary predictor first estimates a sensitive column, then a corrector adjusts it to satisfy a fairness relationship. The strong baseline A' already uses non-sensitive features, true labels, auxiliary data and model predictions. With ThresholdOptimizer post-processing and statistical parity on Adult, accuracy rises from 0.814 to 0.858. Equal opportunity changes it from 0.805 to 0.807 (Table III). The gain depends on which records the relationship covers. Using estimated fairness relationships brings benefits in some settings but also greater variability. These tests cover the chosen estimation procedure; they do not establish robustness to arbitrary incorrect constraints.

These studies motivate two separate comparisons. Comparing DRAFT with and without a known column measures the effect of knowledge at a fixed release. Comparing it with a model-free predictor given the same prior measures the model's added value over that predictor. The interpretation depends on the baseline chosen. Auxiliary training data must be declared, disjoint from the private training table and equally available to the compared conditions.

3.3 Data regimes and the meaning of successful recovery

Adult and COMPAS recur across tree studies, though preprocessing, training sizes and tasks differ. DRAFT and probabilistic reconstruction use different binarizations and training sizes. DRAFT scores an encoded anonymous table, while fairness correction scores one column of known records. Jayaraman and Evans use Texas-100X and Census19. Fredrikson's tree experiments use FiveThirtyEight and GSS. The SoK benchmark [3] mainly uses CelebA, CIFAR10 and MNIST and does not test constraint-based RF reconstruction. It contributes an original comparison of ten attacks and shows why distributional similarity, sample fidelity and coverage need separate measures.

Wen et al. [3] distinguish a feature-space distributional distance, reconstructed-sample nearest-neighbor distances and coverage of distinct originals (Section 4). A small distributional distance can hide differences between records, while nearest-neighbor matching can repeatedly select the same original. Coverage, in turn, says little about distance. These independent assignments can be many-to-one, while DRAFT uses a minimum-cost bijection. Image distances and numerical results apply to the vision setting. For tabular RFs, the useful principle is to state what each metric measures and how reconstructed and original records are matched.

Other tree studies show why the choice of metric matters. In probabilistic reconstruction, the entropy ratio measures remaining uncertainty over compatible tables [2], rather than the proportion of original records recovered. Its comparison of optimal and greedy learners depends on whether model size or training accuracy is fixed. Fredrikson's white-box tree inverter can reach 100% precision on a selected subset, with recall of 21.1% on FiveThirtyEight and 0.7% on GSS. Overall GSS accuracy is 80.3%, below the 82.0% majority baseline [1]. Precision, predictability and coverage should be considered together. The marginal baseline is also weaker than conditional imputation with comparable auxiliary information.

DRAFT's standard matching raises a further issue when the target T is fixed. The evaluator aligns rows using all true feature values, including T, before scoring reconstruction. This is a valid table-distance protocol, but it may improve the target score using evaluator-only information. The score alone cannot show whether the attacker can link a value to a person. Excluding T from the matching cost tests sensitivity to this choice, though the check still uses ground truth for other columns that may be unavailable to the attacker. Repeated known values create ties; these must be resolved without the true T.

Matched bit error, original-category error, exact-row recovery, uncertainty, top-100 PPV and coverage measure different outcomes. Comparisons should fix the target and protocol, report variability and failed solves, and remain specific to each setting.

4 GAPS AND OPEN PROBLEMS

The main RF experiments do not isolate feature identity. The known-column experiments change the number of known attributes and score those that remain unknown [4], [5]. Mehnaz already studies attribute identity in a black-box setting [6]. In the reviewed joint RF studies, the missing comparison is between two known original features, with the target, released forest and solver budget kept fixed. DRAFT's support for known columns makes this question testable. The contrasting findings motivate the comparison, even though they come from different configurations. Given the search's limited coverage, this gap can only be claimed within the reviewed corpus.

The model's added value and row matching need separate controls. Population attributes may be predictable without much information from the model [7]. DRAFT already has a random no-model baseline consistent with the schema. An imputation control would also use permitted auxiliary correlations [4]. Matching can improve scores using information unavailable to the attacker [3], [4]. A no-model control measures the model's added value, while excluding the target from matching tests sensitivity to the scoring protocol. The control needs the same prior information and declared auxiliary data. Repeated values and ties also need reporting. Matching without the target remains a score diagnostic; it does not establish personal identification.

One known feature can provide different amounts of information. Columns can differ in cardinality, entropy, association with the target and ability to distinguish rows. Access to exact values for every training row is a strong assumption, so results describe that specific attacker setting. Fairness already studies estimated relationships [8]. It remains to be tested whether the A/B ordering in joint RF reconstruction holds with partial coverage, errors or distribution shift. Each extension needs a controlled test following the initial exact-column comparison.

Computational feasibility and protocol details need clear reporting. DRAFT provides code and reports failed solves, but its mean error is conditional on finding a solution within the budget [4]. The SoK places some benchmark details in extended material [3], which is needed to repeat the experiments. A reproducible evaluation should report seeds, encoding, release, priors, budgets, solver status, timeouts and paired variability.

5 CONCLUSION: PROPOSED NICHE

Across individual inference, joint reconstruction and uncertainty, the reviewed evidence points to a focused comparison: how much a known column helps, what the model adds and how row matching affects the score.

Proposed question: For a fixed bagged non-DP RF, target feature T and DRAFT budget, how does knowing original feature A rather than B change reconstruction error on T?

The comparison needs a condition with no exact known column, a model-free baseline with the same prior information, and a check of sensitivity to row matching.

Choose distinct original features A, B and T before running the attacks, treating all encoding bits of each feature together. Use the same training sample and forest for three conditions: no exact column, A known for every row, and B known for every row. Keep schema information, label availability, hyperparameters, seeds and solver settings fixed. Measure error on the original category of T. Also report bit error where relevant, paired differences, solver status and runtime. Record each known feature's association with the target, cardinality and row ambiguity, since different columns can provide different amounts of information.

Separate the effect of knowledge from the model's contribution using two comparisons:

$$\Delta_A = \mathrm{Err}_T(\mathrm{DRAFT}, \emptyset) - \mathrm{Err}_T(\mathrm{DRAFT}, A),$$
$$G_A = \mathrm{Err}_T(\text{no model}, A) - \mathrm{Err}_T(\mathrm{DRAFT}, A).$$

Compute the same quantities for B. In the $\emptyset$ condition, the attacker knows the schema but no exact columns. The model-free predictor receives the same permitted prior, with any auxiliary training sample disjoint from the private table and equally available across conditions. It must return a table with the same schema and use the same matching and target metric. G_A measures advantage over this implemented control, which may perform differently from an optimal predictor.

Compare standard DRAFT matching with a sensitivity check that excludes T from the cost. Resolve ties without the true T and report ambiguous matches. The check can still use ground truth for other columns available only to the evaluator. A null gain, a reversed A/B ordering or failure within budget would also be informative. This proposal extends an existing question about feature identity to joint bagged-forest reconstruction. Later experiments can address DP and imperfect knowledge by controlling these factors separately.

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